

SMARTBRIDGE

Let's Bridge the Gap | SKILL WALLET - A SmartBridge Product

Date	29-09-2026
Team ID	SWTID-2026-7949
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Initialization and Planning Phase

Project Proposal (Proposed Solution) Report

The proposal report outlines the design, implementation, and verification of a Script-Controlled Access Control List (ACL) architecture in ServiceNow. It restricts record access based on field values, enforcing least-privilege security, preventing unauthorized cross-department data exposure, and providing granular CRUD governance.

Project Overview	
Objective	Implement a robust, script-controlled Access Control List (ACL) security framework on ServiceNow custom tables (u_institution_details) that dynamically restricts record visibility based on field values (Branch == 'EEE') while preserving administrative oversight.
Scope	Encompasses custom table design, custom security role creation (bb1 to bb4), implementation of 4 discrete server-side ACLs (Read, Create, Write, Delete), and comprehensive validation across multiple user personas via impersonation.
Problem Statement	
Description & Impact	Default ServiceNow table permissions permit broad record visibility, causing unauthorized exposure of sensitive student data across academic departments (CSE, ECE, EEE). Solving this prevents data leaks, enforces data privacy compliance, and eliminates accidental record modifications.
Proposed Solution	
Approach & Key Features	• Implement Script-Controlled Read ACL combining field condition (Branch is EEE) with script evaluation: gs.hasRole('bb1') and gs.hasRole('admin'). • Implement Create ACL requiring role 'bb2' to govern record insertion. • Implement Write ACL requiring role 'bb3' to govern record modification. • Implement Delete ACL requiring role 'bb4' to restrict record deletion to authorized personnel.

Resource Requirements

Resource Type	Description	Specification / Allocation
Cloud Infrastructure	ServiceNow Personal Developer Instance (PDI)	Washington DC / Xanadu Release Cloud Instance (Dedicated PaaS Sandbox)
Server Hardware	Cloud Host Specifications	Enterprise Multi-Tenant Cloud Architecture (High-availability failover)

Development Tooling	Application Configuration & IDE	ServiceNow Studio, Application Navigator, System Security Module
Scripting Engine	Server-Side Script Execution	ECMAScript 2021 (ES512 Mode), ServiceNow GlideSystem API (gs)
Database & Schema	Data Persistence & Models	Custom table 'u_institution_details' with 7 fields (Auto Number, Reference, Choice, String)
Security Framework	Role-Based Access Control (RBAC)	Elevated Role 'security_admin' + 4 custom roles (bb1, bb2, bb3, bb4)
Test Persona Data	Mock User & Student Data Records	1 Test User (EEE User) + 6 Multi-Department Institutional Records (ECE, CSE, EEE)